

Key terms used throughout this chapter

analysis – part of the program development life cycle; a process of investigation, leading to the specification of what a program is required to do

design – part of the program development life cycle; uses the program specification from the analysis stage to show to how the program should be developed

coding – part of the program development life cycle; the writing of the program or suite of programs

testing – part of the program development life cycle; systematic checks done on a program to make sure that it works under all conditions

abstraction – a method used in the analysis stage of the program development life cycle; the key elements required for the solution to the problem are kept and any unnecessary details and information that are not required are discarded

decomposition – a method used in the analysis stage of the program development life cycle; a complex problem is broken down into smaller parts, which can then be sub divided into even smaller parts that can be solved more easily

top-down design – the breaking down of a computer system into a set of sub-systems, then breaking each sub-system down into a set of smaller sub-systems, until each sub-system just performs a single action

inputs – the data used by the system that needs to be entered while the system is active

processes – the tasks that need to be performed by a program using the input data and any other previously stored data

output – information that needs to be displayed or printed for the users of the system

storage – data that needs to be stored in files on an appropriate media for use in the future

structure diagram – a diagram that shows the design of a computer system in a hierarchical way, with each level giving a more detailed breakdown of the system into sub-systems

flowchart – a diagram that shows the steps required for a task (sub-system) and the order in which the steps are to be performed

algorithm – an ordered set of steps to solve a problem

pseudocode – a simple method of showing an algorithm; it describes what the algorithm does by using English key words that are very similar to those used in a high-level programming language but without the strict syntax rules

linear search – an algorithm that inspects each item in a list in turn to see if the item matches the value searched for

bubble sort – an algorithm that makes multiple passes through a list comparing each element with the next element and swapping them. This continues until there is a pass where no more swaps are made

validation – automated checks carried out by a program that data is reasonable before it is accepted into a computer system

verification – checking that data has been accurately copied from another source and input into a computer or transferred from one part of a computer system to another

set of test data – all the items of data required to work through a solution

normal data – data that is accepted by a program

abnormal data – data that is rejected by a program

extreme data – the largest/smallest data value that is accepted by a program

boundary data – the largest/smallest data value that is accepted by a program and the corresponding smallest/largest rejected data value

range check – a check that the value of a number is between an upper value and a lower value

length check – a method used to check that the data entered is a specific number of characters long or that the number of characters is between an upper value and a lower value

type check – a check that the data entered is of a specific type

presence check – a check that a data item has been entered

format check – a check that the characters entered conform to a pre-defined pattern

check digit – an additional digit appended to a number to check if the entered number is error-free; check digit is a data entry check and not a data transmission check